

**BNs: Marginalisation**

$X_1 \rightarrow X_2 \rightarrow X_3 \rightarrow X_4$

$P(X_2) = P(X_1)P(X_2|X_1)$   
 $P(X_3|X_2) = P(X_1)P(X_2|X_1)P(X_3|X_2)$   
 $P(X_4|X_3) = P(X_1)P(X_2|X_1)P(X_3|X_2)P(X_4|X_3)$

**MCs: Assumption** (first order)

1. MCs (of order n)  
Pr of outcome depends on the previous n states

2. Stationarity (Assumed)  
Transition Prs same

**MC in NLP**

Unigram  $w_1, w_2, w_3$   
Bigram  $w_1 \rightarrow w_2 \rightarrow w_3$   
Trigram  $w_1 \rightarrow w_2 \rightarrow w_3$

$P(w_1, w_2) = P(w_1)P(w_2|w_1)$   
 $P(w_3) | w_1, w_2$

**MCs are BNs**

Semantics apply.

Local: Cond. indep.  $X \perp Y | Z$   
 $P(X|Y, Z) = P(X|Z)$   
 $P(X|Z) = P(Y|Z)$  Markov blanket: parents + children + children's parents.

d-separation: (C, A, B, M)  
 $X \rightarrow Z \rightarrow Y$  of apply Z.  
 $X \perp Y | Z$

**Joint distr.**

$X_1 \rightarrow X_2 \rightarrow \dots \rightarrow X_t$   
 $P(\dots) = P(X_1) \prod_{i=1}^{t-1} P(X_{i+1}|X_i)$

Thus only need  $X_1$  tra. pr.

(Cond. prob table, CPT)

**Representation e.g. Stock**

$V_1 \rightarrow V_2 \rightarrow \dots$

OCPT  
 $V_{t-1} \in \mathbb{R}^d, P(V_t|V_{t-1})$   
 $T = \begin{bmatrix} h_{t-1} & u_t \\ h_t & u_{t+1} \end{bmatrix}$

State Tr. Matrix  
 $T = \begin{bmatrix} h_{t-1} & u_t \\ h_t & u_{t+1} \end{bmatrix}$

State Tr. Diagram  
 $h_{t-1} \rightarrow h_t$   
 $u_t \rightarrow h_t$   
 $u_t \rightarrow u_{t+1}$

**Prediction**

$P(X_t) = TP(X_{t-1})$   
 $= \dots = T^{t-1} P(X_1)$

Stationary distr.  
 $P(X_t) = P(X_1)$

$P(X_t) = P(X_{t-1})$   
 $= T^{t-1} P(X_1)$   
ie. eigen of  $T$

**Example**

$P_{\text{low}}(\text{high}) = P(H|H)P(H)$   
 $P(H|H) = \text{osp}(H)$   
 $T = 0.1 P(H), P(H|\text{low})$   
 $= 1 - P_{\text{low}}(\text{high})$   
 $(Z, P_{\text{low}}(U) = 1)$

**Application**

HMM as BNN by factoring add. evidence

Diffusion Model  
range rank  
Tij = P(Xt=i) = 1 - P(Xt=j)

HMM  
 $X_t$ : State,  $E_t$ : observation  
 $X_1 \rightarrow X_2 \rightarrow X_3$ : hidden/unobserved states  
 $E_1, E_2, E_3$ : evidence/emissions

**HMM stock ex.**

S: stock value  
 $XES = \{h, e\}$   
O: Buy or sell  
 $EEO = \{b, s\}$   
 $X_1 \rightarrow X_2 \rightarrow \dots$   
 $E_1, E_2, \dots$

**Joint pd.**

$P(E_1, \dots, E_t, X_1, \dots, X_t)$   
 $= P(X_1) \prod_{i=1}^{t-1} P(X_{i+1}|X_i) \prod_{i=1}^t P(E_i|X_i)$

init. prob.  $P(X_1)$   
trans. distr.  $P(X_{i+1}|X_i)$   
emission distr.  $P(E_i|X_i)$

**Interacting**

Filtering:  $P(X_t|E_{1:t})$   
Prediction:  $P(X_{t+1}|E_{1:t})$   
Smoothing:  $P(X_t|E_{1:t+1})$   
Most likely explanation:  $P(X_{1:t}|E_{1:t+1})$   
avg. value  $P(X_t|E_{1:t+1})$   
(Viterbi)

**Bayesian Recursive Filtering**

Calculate  $B(X_t) = P(X_t|E_{1:t})$

i) Predict  $B(X_{t+1}) = P(X_{t+1}|E_{1:t})$   
ii) Update with  $E_{t+1}$   
 $B(X_{t+1}|E_{1:t+1}) = \alpha P(E_{t+1}|X_{t+1}) B(X_{t+1}|E_{1:t})$

**Logistic (cts.)**

NOT MSE: non-convex!  
Threshold:  $h(x) = \begin{cases} 1, & w^T x > 0 \\ 0, & \text{otherwise} \end{cases}$   
HCP(X) =  $-\sum P(X=x) \log P(X=x)$   
Sigmoid (Logistic):  $h(x) = \frac{1}{1 + e^{-w^T x}}$   
Cross-entropy:  $H(P_{\text{true}}, P_{\text{pred}}) = -\sum P_{\text{true}}(c) \log P_{\text{pred}}(c)$

**Lin. model**

$x^{(i)} = \begin{bmatrix} x_1^{(i)} \\ x_2^{(i)} \end{bmatrix}$   
 $\hat{y} = g(w^T x^{(i)})$   
 $w = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \end{bmatrix}$  weight

**Lin. reg.** ( $L_2: \|w\|_2^2$ )

$h_w(x) = w^T x$   
 $w = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \end{bmatrix}$   
MSE:  $J(w) = \frac{1}{2m} \sum_{i=1}^m (h_w(x^{(i)}) - y^{(i)})^2$   
Batch: all data  
SGD: checkerboard mini-batch

**Cost**

$\text{Cost}(h_w(x), y) = -\log(h_w(x), y) = -\log(h_w(x))$   
 $y = 1$   
 $y = 0$   
 $y = 1$ :  $h_w(x) \rightarrow 1, J(w) \rightarrow 0$   
 $y = 0$ :  $h_w(x) \rightarrow 0, J(w) \rightarrow \infty$

**Multi-class: pick C st.**

$\max_c h_w^{(c)}(x)$   
softmax( $\vec{x}$ ) =  $\frac{e^{x_i}}{\sum e^{x_j}}$   
Recall =  $\frac{\text{TP}}{\text{TP} + \text{FN}}$   
Precision =  $\frac{\text{TP}}{\text{TP} + \text{FP}}$   
F1 =  $\frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$

**MLP**

$z^{(L)} = W^{(L)} a^{(L-1)} + b^{(L)}$   
 $a^{(L)} = g(z^{(L)})$   
Backprop:  $\delta^{(L)} = \frac{\partial \text{Loss}}{\partial z^{(L)}}$   
Output:  $\delta^{(L)} = \frac{\partial \text{Loss}}{\partial z^{(L)}}$   
Hidden:  $\delta^{(L)} = (W^{(L+1)} \delta^{(L+1)}) \odot g'(z^{(L)})$   
# of param:  $N_L(N_L + 1)$

**Losses**

Huber etc. more smooth, not less sensitive. robust

RCE  $J(w) = H(P(y=1|w), P(y=0|w))$   
 $= -\sum_{c \in \text{class}} P(y=c) \log(P(y=c))$   
 $= \text{negative log-likelihood!}$   
 $J(w) = \frac{1}{m} \sum_{i=1}^m \text{Cost}(h_w(x^{(i)}), y^{(i)})$

**Normal equation**

$W = (X^T X)^{-1} X^T y$   
Non-invertible: adding 1s  
- lin. dep.  
- features > samples  
What to do?  
- remove features  
- regularization (ridge/L2)

**Losses II**

MSE: sensitive to outliers  
MAE: robust, non-smooth  
Huber/Loss: MAE, MSE: Luse, MAE: Luse, MAE: Luse, MAE: Luse  
Thus, LFP: pre. (spem. phys.)  
L2F: rec. (linear, dense)

**Perception**

$\hat{y} = \text{sign}(\sum w_i x_i)$   
RLA: - initial  $w_i = 0$   
-  $\hat{y}_i = \text{sign}(w_i x_i)$   
-  $w_i \leftarrow w_i + \Delta w$   
- converge?  
Learning rate  $\leftarrow \gamma(y - \hat{y})x$

**Linearly Separable it**

$\exists w: w^T x > 0 \Rightarrow P$   
 $\exists$ : converge?  
 $\exists$ : oscillate forever.  
non-linear  $\rightarrow$  MLP.

**Back Propagation: Scalar Example**

At each layer, we propagate gradients back to the input.

Layer 1:  $x_1, x_2$   
Layer 2:  $z_1, z_2$   
Layer 3:  $a_1, a_2$   
Output:  $y$

Loss function:  $J = \frac{1}{2} (y - \hat{y})^2$

Gradients approach zero as number of layers increases

**Vanishing Gradient Problem**

Gradients approach zero as number of layers increases

Layer 1:  $x_1, x_2$   
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**Back Propagation: Vector Example**

At each layer, we propagate gradients back to the input.

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Output:  $y$

Loss function:  $J = \frac{1}{2} (y - \hat{y})^2$

**Gradients in MLP: Intuition**

$\delta^{(L)} = g'(z^{(L)}) (y - \hat{y})$   
 $\delta^{(L-1)} = (W^{(L)} \delta^{(L)}) \odot g'(z^{(L-1)})$   
Recursion

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| Activation Function                                                                                   | Range               | Use case                          |
|-------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------|
| Identity: $g(x) = x$                                                                                  | $[-\infty, \infty]$ | Linear Regression                 |
| ReLU: $g(x) = \begin{cases} x & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$              | $[0, \infty]$       | Linear Regression, Classification |
| Leaky ReLU: $g(x) = \begin{cases} x & \text{if } x \geq 0 \\ \alpha x & \text{if } x < 0 \end{cases}$ | $[-\infty, \infty]$ | Linear Regression, Classification |
| Sigmoid: $g(x) = \frac{1}{1 + e^{-x}}$                                                                | $(0, 1)$            | Linear Regression, Classification |
| Tanh: $g(x) = \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$                                           | $(-1, 1)$           | Linear Regression, Classification |
| Softmax: $g(x) = \frac{e^{x_i}}{\sum_j e^{x_j}}$                                                      | $(0, 1)$            | Multiclass Classification         |

| Loss Function                 | Formula                                                                                                                                                                           | Use case                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| Mean Square Error             | $J(w) = \frac{1}{2} \sum_{i=1}^n (y_i - \hat{y}_i)^2$                                                                                                                             | Linear Regression                        |
| Mean Absolute Error           | $J(w) = \frac{1}{n} \sum_{i=1}^n  y_i - \hat{y}_i $                                                                                                                               | Linear Regression                        |
| Huber Loss                    | $J(w) = \begin{cases} \frac{1}{2} (y_i - \hat{y}_i)^2 & \text{if }  y_i - \hat{y}_i  \leq \delta \\ \delta  y_i - \hat{y}_i  - \frac{\delta^2}{2} & \text{otherwise} \end{cases}$ | Linear Regression                        |
| Tanh-Cross Loss               | $J(w) = \frac{1}{n} \sum_{i=1}^n \log(\cosh(y_i - \hat{y}_i))$                                                                                                                    | Linear Regression                        |
| Binary Cross Entropy Loss     | $J(w) = -\frac{1}{n} \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$                                                                                          | Linear Regression, Binary Classification |
| Multiclass Cross Entropy Loss | $J(w) = -\frac{1}{n} \sum_{i=1}^n \sum_{j=1}^K y_{ij} \log(\hat{y}_{ij})$                                                                                                         | Multiclass Classification                |
| Hinge Loss                    | $J(w) = \frac{1}{n} \sum_{i=1}^n \max(0, 1 - y_i \hat{y}_i)$                                                                                                                      | Binary Classification                    |

| Optimizers                      | Update Rule                                                                                                                                                                              | Characteristics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           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| Adaptive Gradient Descent (AGD) | $\theta_t = \theta_{t-1} + \frac{\eta}{\sqrt{G_t}} \nabla J(\theta_{t-1})$                                                                                                               | <ul style="list-style-type: none"><li>Adaptive learning rate for all weights to converge fast</li><li>Use learning rate <math>\eta</math> at all weights to converge for convergence</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            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| Adam                            | $\begin{aligned} m_t &= m_{t-1} + \frac{\eta}{\sqrt{\hat{G}_t}} \nabla J(\theta_{t-1}) \\ \theta_t &= \theta_{t-1} + \frac{\eta}{\sqrt{\hat{G}_t}} \nabla J(\theta_{t-1}) \end{aligned}$ | <ul style="list-style-type: none"><li>Use learning rate <math>\eta</math> for all weights</li><li>Use learning rate <math>\eta</math> for all weights to converge for convergence</li><li>Use learning rate <math>\eta</math> for all weights to converge for convergence</li><li>Use learning rate <math>\eta</math> for all weights to converge for convergence</li><li>Use learning rate <math>\eta</math> for all weights to converge for convergence</li><li>Use learning rate <math>\eta</math> for all weights to converge for convergence</li><li>Use learning rate <math>\eta</math> for all weights to converge for convergence</li><li>Use learning rate <math>\eta</math> for all weights to converge for convergence</li><li>Use learning rate <math>\eta</math> for all weights to converge 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